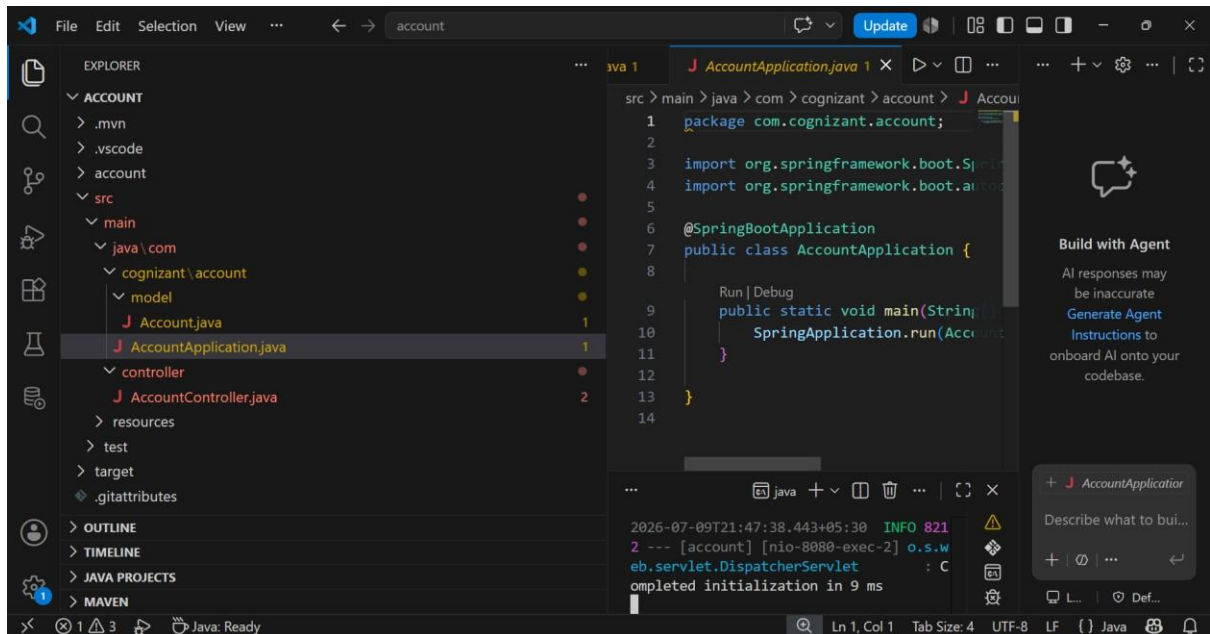


DIGITAL NURTURE 5.0 DEEPSKILLING

WEEK 4 – HANDS-ON

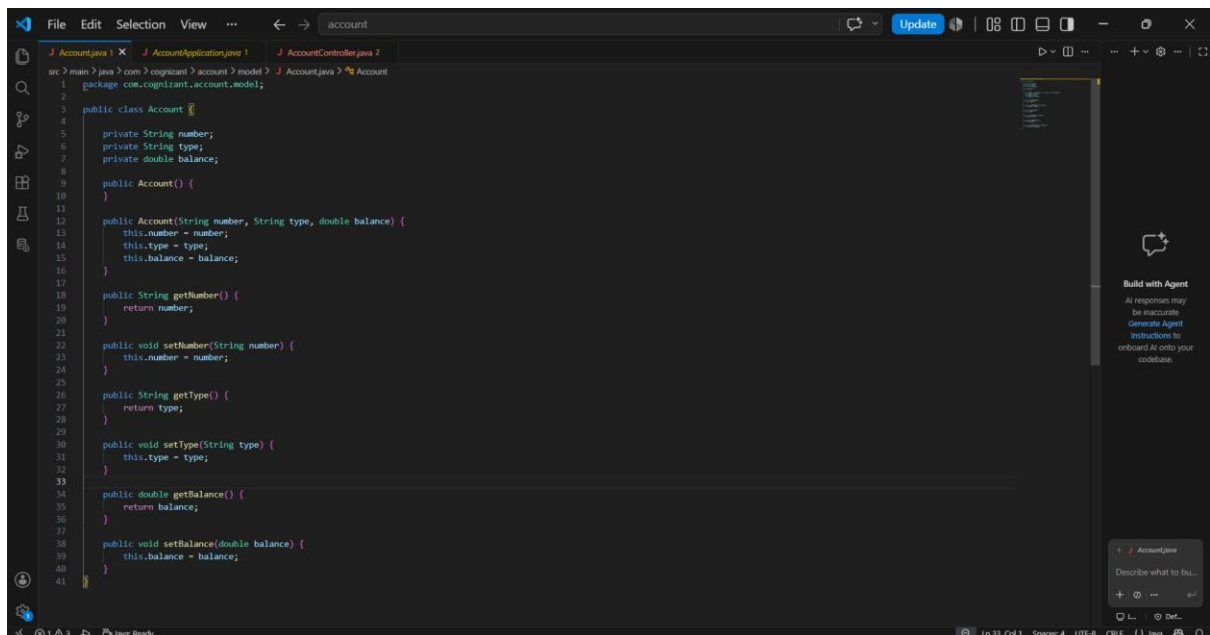
Creating Microservices for account and loan

ACCOUNT :



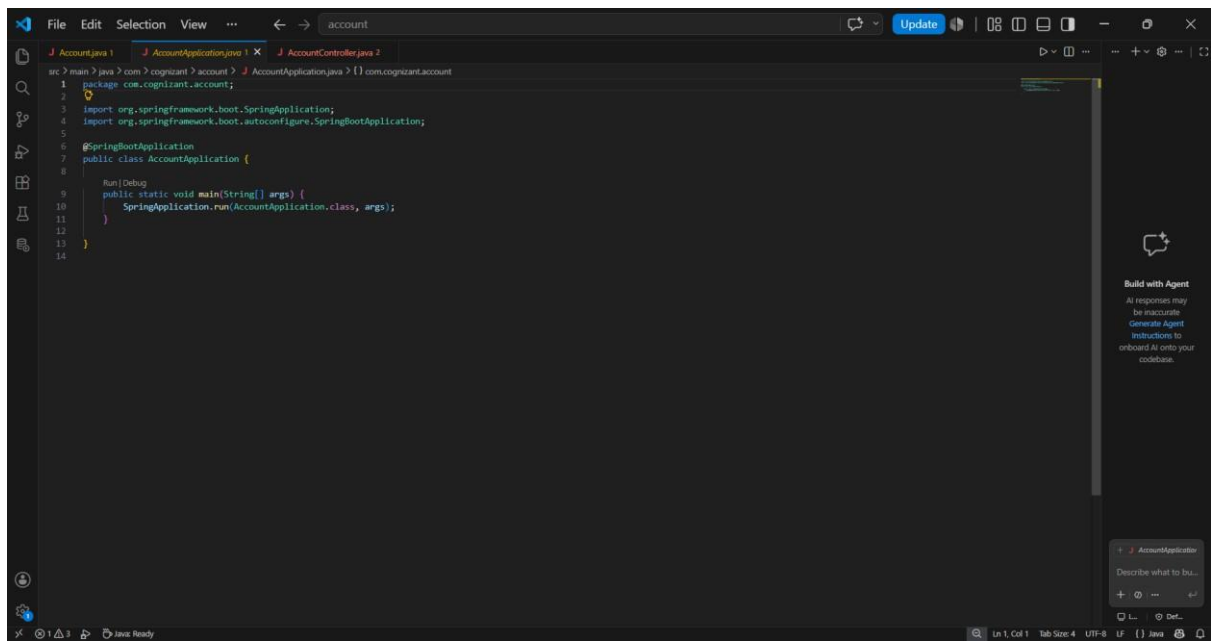
The screenshot shows the Visual Studio Code editor with the 'account' project open. The Explorer panel on the left shows the project structure: `ACCOUNT` (root) contains `.mvn`, `.vscode`, `account`, `src` (with `main` and `test` subfolders), `resources`, `target`, and `.gitattributes`. The `src/main/java/com/cognizant/account` package contains `Account.java`, `AccountApplication.java` (selected), and `AccountController.java`. The main editor displays the `AccountApplication.java` file, which is a Spring Boot application. The code includes package declarations, imports for Spring Boot and Spring Web, and a `@SpringBootApplication` annotation. The `main` method is defined as `public static void main(String[] args) { SpringApplication.run(AccountApplication.class, args); }`. The status bar at the bottom indicates 'Ln 1, Col 1' and 'Tab Size: 4'.

```
src > main > java > com > cognizant > account > J AccountApplication.java 1 x
1 package com.cognizant.account;
2
3 import org.springframework.boot.SpringApplication;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6 @SpringBootApplication
7 public class AccountApplication {
8
9     Run | Debug
10    public static void main(String[] args) {
11        SpringApplication.run(AccountApplication.class, args);
12    }
13 }
14
```



The screenshot shows the Visual Studio Code editor with the 'account' project open. The Explorer panel on the left shows the project structure, with `Account.java` selected in the `src/main/java/com/cognizant/account` package. The main editor displays the `Account.java` file, which is a Java class representing an account. The code includes package declarations, private fields for `number`, `type`, and `balance`, and methods for `getNumber`, `setNumber`, `getType`, `setType`, `getBalance`, and `setBalance`. The status bar at the bottom indicates 'Ln 33, Col 1' and 'Tab Size: 4'.

```
src > main > java > com > cognizant > account > model > J Account.java 2
1 package com.cognizant.account.model;
2
3 public class Account {
4
5     private String number;
6     private String type;
7     private double balance;
8
9     public Account() {
10    }
11
12    public Account(String number, String type, double balance) {
13        this.number = number;
14        this.type = type;
15        this.balance = balance;
16    }
17
18    public String getNumber() {
19        return number;
20    }
21
22    public void setNumber(String number) {
23        this.number = number;
24    }
25
26    public String getType() {
27        return type;
28    }
29
30    public void setType(String type) {
31        this.type = type;
32    }
33
34    public double getBalance() {
35        return balance;
36    }
37
38    public void setBalance(double balance) {
39        this.balance = balance;
40    }
41 }
```



```
File Edit Selection View ... account
J Account.java 1 J AccountApplication.java 1 J AccountController.java 2 X
src > main > java > com > controller > J AccountController.java > AccountController
1 package com.cognizant.account.controller;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.PathVariable;
5 import org.springframework.web.bind.annotation.RestController;
6
7 import com.cognizant.account.model.Account;
8
9 @RestController
10 public class AccountController {}
11
12 @GetMapping("/accounts/{number}")
13 public Account getAccount(@PathVariable String number) {
14     return new Account(number, type: "Savings", balance: 234343);
15 }
16
17
18
```

```
File Edit Selection View ... account
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SQL HISTORY TASK MONITOR
Microsoft Windows [Version 10.0.26000.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Microservices\account>mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----[ jar ]-----
[INFO] Building 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ account ---
[INFO] Deleting C:\Microservices\account\target
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ account ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ account ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug parameters release 17] to target/classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ account ---
[INFO] skip non existing resourceDirectory C:\Microservices\account\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ account ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug parameters release 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.5.6:test (default-test) @ account ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -----[ jar ]-----
[INFO] T E S T S
[INFO]
[INFO] Running com.cognizant.account.AccountApplicationTests
21:46:04.513 [main] INFO org.springframework.test.context.support.AnnotationConfigContextLoaderUtils -- Could not detect default configuration classes for test class [com.cognizant.account.AccountApplicationTests]: AccountApplicationTests does not declare any static, non-private, non-final, nested classes annotated with @Configuration.
21:46:04.774 [main] INFO org.springframework.boot.test.context.SpringBootTestContextBootstrapper -- Found @SpringBootConfiguration com.cognizant.account.AccountApplication for test class com.cognizant.account.AccountApplicationTests
21:46:04.927 [main] INFO org.springframework.test.context.support.AnnotationConfigContextLoaderUtils -- Could not detect default configuration classes for test class [com.cognizant.account.AccountApplicationTests]: AccountApplicationTests does not declare any static, non-private, non-final, nested classes annotated with @Configuration.
21:46:04.928 [main] INFO org.springframework.boot.test.context.SpringBootTestContextBootstrapper -- Found @SpringBootConfiguration com.cognizant.account.AccountApplication for test class com.cognizant.account.AccountApplicationTests
21:46:05.017 [main] INFO org.springframework.boot.devtools.restart.RestartApplicationListener -- Restart disabled due to context in which it is running
```

```
File Edit Selection View ... account Update
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SQL HISTORY TASK MONITOR

:: Spring Boot :: (v4.1.0)

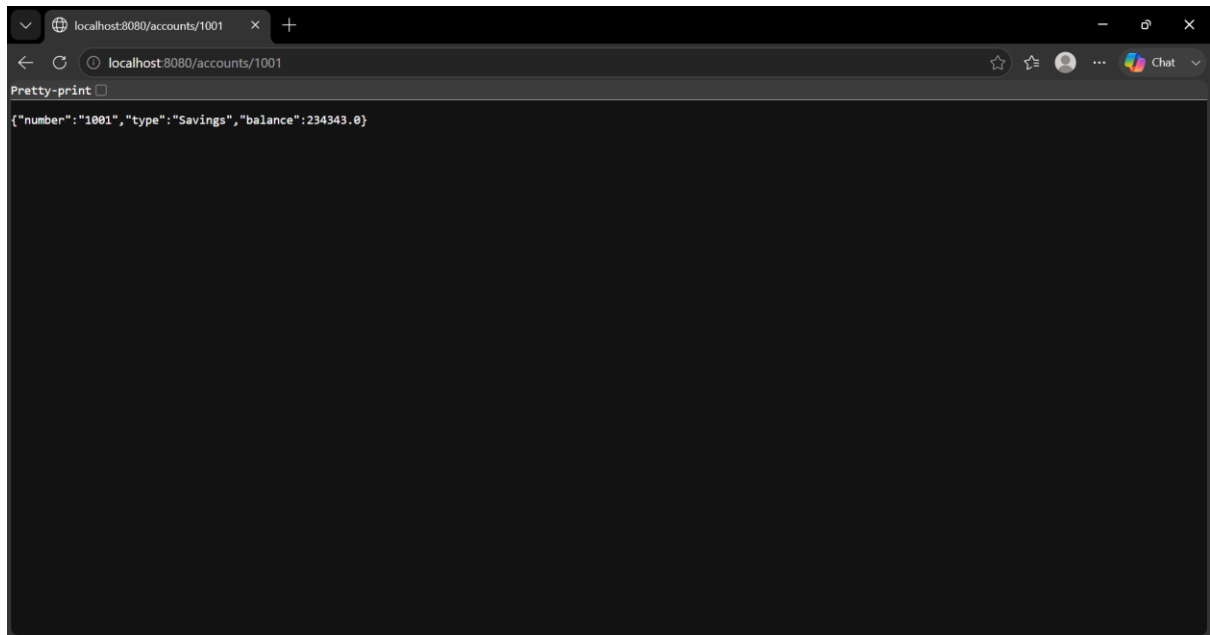
2026-07-09T21:46:05.352+05:30 INFO 15608 --- [account] [ main] c.c.account.AccountApplicationTests : Starting AccountApplicationTests using Java 21.0.2 with PID 15608 (started by srini in C:\Microservices\account)
2026-07-09T21:46:05.354+05:30 INFO 15608 --- [account] [ main] c.c.account.AccountApplicationTests : No active profile set, falling back to 1 default profile: "default"
2026-07-09T21:46:06.602+05:30 INFO 15608 --- [account] [ main] c.c.account.AccountApplicationTests : Started AccountApplicationTests in 1.603 seconds (process running for 3.087)
Mockito is currently self-attaching to enable the inline mock-maker. This will no longer work in future releases of the JDK. Please add Mockito as an agent to your build as described in Mockito's documentation: https://www.mockito.org/docs/org.mockito.core/jar/test/org.mockito/org.mockito.html#0.2
WARNING: A Java agent has been loaded dynamically (C:\Users\srini\.m2\repository\net\bytebuddy\byte-buddy-agent\1.18.10\byte-buddy-agent-1.18.10.jar)
WARNING: If a serviceability tool is in use, please run with -XX:EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 3.106 s -- in com.cognizant.account.AccountApplicationTests
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ account ---
[INFO] Building jar: C:\Microservices\account\target\account-0.0.1-SNAPSHOT.jar
[INFO]
[INFO] --- spring-boot-docker-plugin:3.0.0:repackage (repackage) @ account ---
[INFO] Replacing main artifact C:\Microservices\account\target\account-0.0.1-SNAPSHOT.jar with repackaged archive, adding nested dependencies in BOOT-INF/.
[INFO] The original artifact has been renamed to C:\Microservices\account\target\account-0.0.1-SNAPSHOT.jar.original
[INFO]
[INFO] --- install:3.1.4:install (default-install) @ account ---
[INFO] Installing C:\Microservices\account\target\pom.xml to C:\Users\srini\.m2\repository\com\cognizant\account\0.0.1-SNAPSHOT\account-0.0.1-SNAPSHOT.pom
[INFO] Installing C:\Microservices\account\target\account-0.0.1-SNAPSHOT.jar to C:\Users\srini\.m2\repository\com\cognizant\account\0.0.1-SNAPSHOT\account-0.0.1-SNAPSHOT.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 9.345 s
[INFO] Finished at: 2026-07-09T21:46:09+05:30
[INFO]
C:\Microservices\account>mvn spring-boot:run
[INFO] Scanning for projects...
[INFO]
[INFO] ----- com.cognizant:account >-----
[INFO] Building 0.0.1-SNAPSHOT
[INFO]
```

```
File Edit Selection View ... account Update
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SQL HISTORY TASK MONITOR

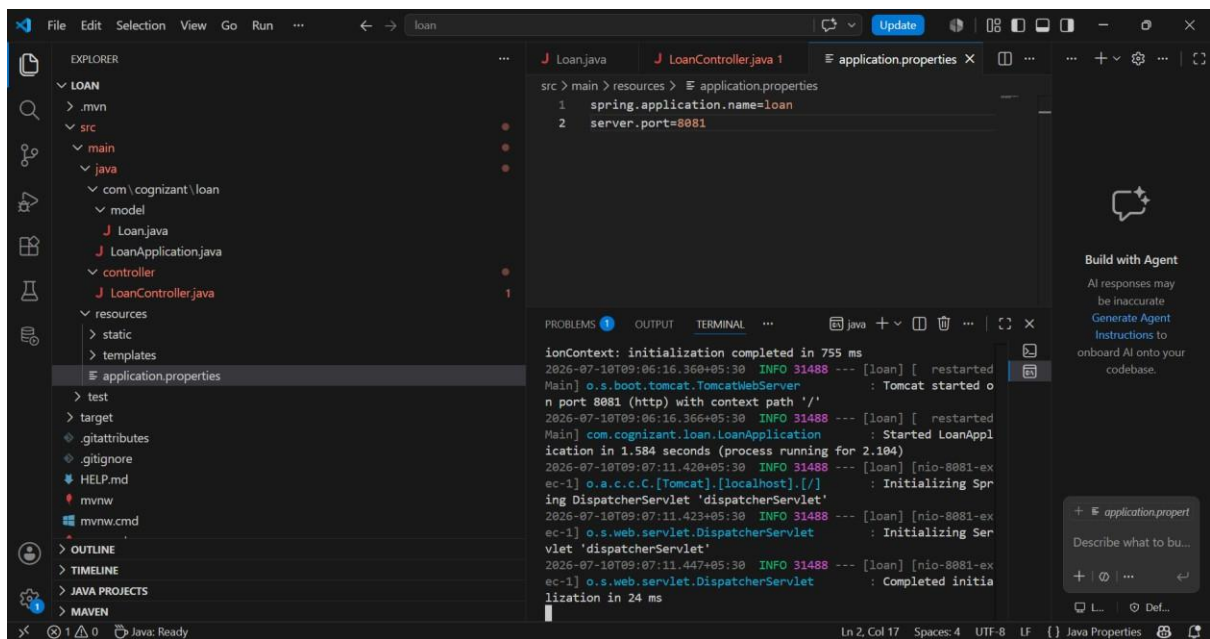
[INFO] Building 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO]
[INFO] -----[ jar ]-----
[INFO]
[INFO] >>> spring-boot:4.1.0:run (default-cli) < test-compile @ account >>>
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ account ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ account ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug parameters release 17] to target/classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ account ---
[INFO] skip non existing resourceDirectory C:\Microservices\account\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ account ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug parameters release 17] to target\test-classes
[INFO]
[INFO] <<< spring-boot:4.1.0:run (default-cli) < test-compile @ account <<<
[INFO]
[INFO]
[INFO] --- spring-boot:4.1.0:run (default-cli) @ account ---
[INFO] Attaching agents: []

:: Spring Boot :: (v4.1.0)

2026-07-09T21:46:45.876+05:30 INFO 8212 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : Starting AccountApplication using Java 21.0.2 with PID 8212 (C:\Microservices\account\target\classes started by srini in C:\Microservices\account)
2026-07-09T21:46:45.879+05:30 INFO 8212 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-09T21:46:45.938+05:30 INFO 8212 --- [account] [ restartedMain] o.e.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
2026-07-09T21:46:45.939+05:30 INFO 8212 --- [account] [ restartedMain] o.e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property to 'DEBUG'
2026-07-09T21:46:46.568+05:30 INFO 8212 --- [account] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-09T21:46:46.580+05:30 INFO 8212 --- [account] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-09T21:46:46.607+05:30 INFO 8212 --- [account] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-09T21:46:46.687+05:30 INFO 8212 --- [account] [ restartedMain] b.w.c.c.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 667 ms
2026-07-09T21:46:46.880+05:30 INFO 8212 --- [account] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-07-09T21:46:46.884+05:30 INFO 8212 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : Started AccountApplication in 1.385 seconds (process running for 1.721)
```



LOAN :



The screenshot shows the Visual Studio Code editor with the file `Loan.java` open. The code defines a `Loan` class with the following structure:

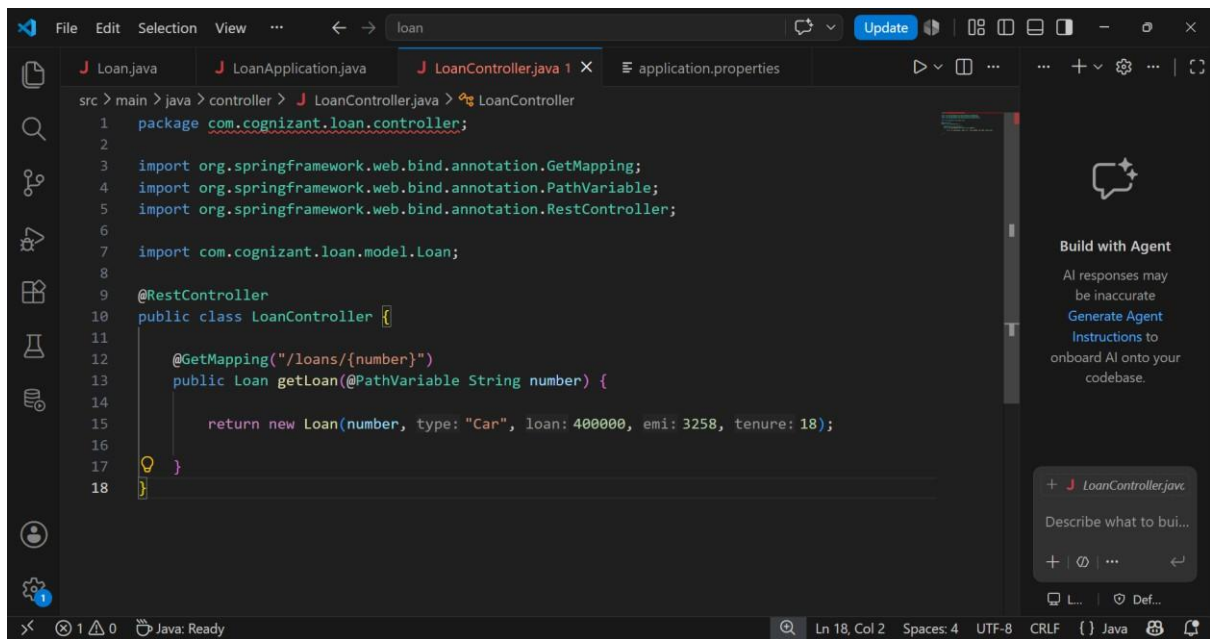
```
src > main > java > com > cognizant > loan > model > Loan.java > Loan
1 package com.cognizant.loan.model;
2
3 public class Loan {
4     private String number;
5     private String type;
6     private double loan;
7     private int emi;
8     private int tenure;
9
10    public Loan() {}
11
12    public Loan(String number, String type, double loan, int emi, int tenure) {
13        this.number = number;
14        this.type = type;
15        this.loan = loan;
16        this.emi = emi;
17        this.tenure = tenure;
18    }
19
20    public String getNumber() {
21        return number;
22    }
23
24    public String getType() {
25        return type;
26    }
27
28    public double getloan() {
29        return loan;
30    }
31
32    public int getemi() {
33        return emi;
34    }
35
36    public int gettenure() {
37        return tenure;
38    }
39
40 }
```

The right sidebar contains the "Build with Agent" panel with a warning that AI responses may be inaccurate and a "Generate Agent" button. Below it is a chat input field with the text "Describe what to bu...".

The screenshot shows the Visual Studio Code editor with the file `LoanApplication.java` open. The code defines a `LoanApplication` class with the following structure:

```
src > main > java > com > cognizant > loan > LoanApplication.java > { } com.cognizant.loan
1 package com.cognizant.loan;
2
3 import org.springframework.boot.SpringApplication;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6 @SpringBootApplication
7 public class LoanApplication {
8
9     Run | Debug
10    public static void main(String[] args) {
11        SpringApplication.run(LoanApplication.class, args);
12    }
13
14 }
```

The right sidebar contains the "Build with Agent" panel with a warning that AI responses may be inaccurate and a "Generate Agent" button. Below it is a chat input field with the text "Describe what to bu...".

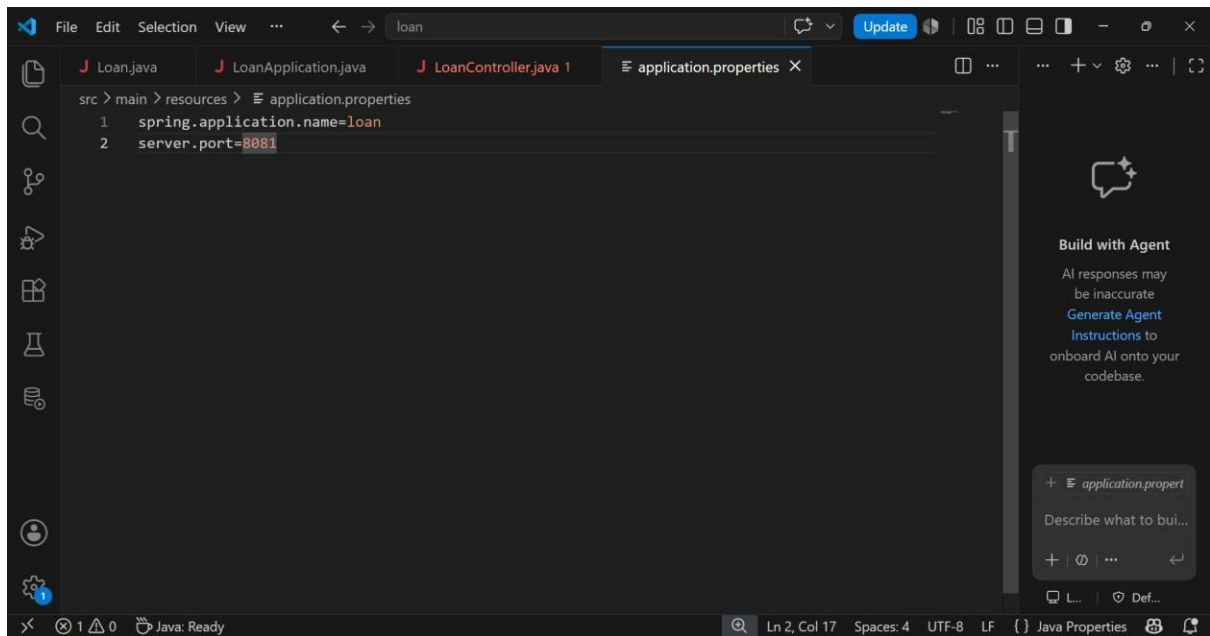


```
src > main > java > controller > J LoanController.java > LoanController
1 package com.cognizant.loan.controller;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.PathVariable;
5 import org.springframework.web.bind.annotation.RestController;
6
7 import com.cognizant.loan.model.Loan;
8
9 @RestController
10 public class LoanController {
11
12     @GetMapping("/loans/{number}")
13     public Loan getLoan(@PathVariable String number) {
14
15         return new Loan(number, type: "Car", loan: 400000, emi: 3258, tenure: 18);
16
17     }
18 }
```

Build with Agent
AI responses may be inaccurate
[Generate Agent Instructions](#) to onboard AI onto your codebase.

+ J LoanController.java
Describe what to bui...
+ | | ... |
| L... | | Def...

Ln 18, Col 2 Spaces: 4 UTF-8 CRLF {} Java



```
src > main > resources > application.properties
1 spring.application.name=loan
2 server.port=8081
```

Build with Agent
AI responses may be inaccurate
[Generate Agent Instructions](#) to onboard AI onto your codebase.

+ application.properties
Describe what to bui...
+ | | ... |
| L... | | Def...

Ln 2, Col 17 Spaces: 4 UTF-8 LF {} Java Properties

```
File Edit Selection View ... loan Update
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SQL HISTORY TASK MONITOR
Microsoft Windows [Version 10.0.26200.855]
(c) Microsoft Corporation. All rights reserved.

C:\Microservices\loan\loan> clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.cognizant:loan -----
[INFO] Building 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ loan ---
[INFO] Deleting C:\Microservices\loan\loan\target
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ loan ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ loan ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug parameters release 17] to target/classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ loan ---
[INFO] skip non existing resourceDirectory C:\Microservices\loan\loan\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ loan ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug parameters release 17] to target/test-classes
[INFO]
[INFO] --- surefire:3.5.6:test (default-test) @ loan ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnit4PlatformProvider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.cognizant.loan.LoanApplicationTests
09:01:42.855 [main] INFO org.springframework.test.context.support.AnnotationConfigContextLoaderUtils -- Could not detect default configuration classes for test class [com.cognizant.loan.LoanApplicationTests]:
LoanApplicationTests does not declare any static, non-private, non-final, nested classes annotated with @Configuration.
09:01:43.055 [main] INFO org.springframework.boot.test.context.SpringBootTestContextBootstrapper -- Found @SpringBootConfiguration com.cognizant.loan.LoanApplicationTests for test class com.cognizant.loan.LoanApplicationTests
09:01:43.321 [main] INFO org.springframework.test.context.support.AnnotationConfigContextLoaderUtils -- Could not detect default configuration classes for test class [com.cognizant.loan.LoanApplicationTests]:
LoanApplicationTests does not declare any static, non-private, non-final, nested classes annotated with @Configuration.
09:01:43.325 [main] INFO org.springframework.boot.test.context.SpringBootTestContextBootstrapper -- Found @SpringBootConfiguration com.cognizant.loan.LoanApplicationTests for test class com.cognizant.loan.LoanApplicationTests
09:01:43.583 [main] INFO org.springframework.boot.devtools.restart.RestartApplicationListener -- Restart disabled due to context in which it is running

Ln 2, Col 17 Spaces: 4 UTF-8 LF Java Properties
```

```
File Edit Selection View ... loan Update
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SQL HISTORY TASK MONITOR
Microsoft Windows [Version 10.0.26200.855]
(c) Microsoft Corporation. All rights reserved.

C:\Microservices\loan\loan> clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.cognizant:loan -----
[INFO] Building 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ loan ---
[INFO] Deleting C:\Microservices\loan\loan\target
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ loan ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ loan ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug parameters release 17] to target/classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ loan ---
[INFO] skip non existing resourceDirectory C:\Microservices\loan\loan\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ loan ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug parameters release 17] to target/test-classes
[INFO]
[INFO] --- surefire:3.5.6:test (default-test) @ loan ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnit4PlatformProvider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.cognizant.loan.LoanApplicationTests
09:01:42.855 [main] INFO org.springframework.test.context.support.AnnotationConfigContextLoaderUtils -- Could not detect default configuration classes for test class [com.cognizant.loan.LoanApplicationTests]:
LoanApplicationTests does not declare any static, non-private, non-final, nested classes annotated with @Configuration.
09:01:43.055 [main] INFO org.springframework.boot.test.context.SpringBootTestContextBootstrapper -- Found @SpringBootConfiguration com.cognizant.loan.LoanApplicationTests for test class com.cognizant.loan.LoanApplicationTests
09:01:43.321 [main] INFO org.springframework.test.context.support.AnnotationConfigContextLoaderUtils -- Could not detect default configuration classes for test class [com.cognizant.loan.LoanApplicationTests]:
LoanApplicationTests does not declare any static, non-private, non-final, nested classes annotated with @Configuration.
09:01:43.325 [main] INFO org.springframework.boot.test.context.SpringBootTestContextBootstrapper -- Found @SpringBootConfiguration com.cognizant.loan.LoanApplicationTests for test class com.cognizant.loan.LoanApplicationTests
09:01:43.583 [main] INFO org.springframework.boot.devtools.restart.RestartApplicationListener -- Restart disabled due to context in which it is running

Ln 2, Col 17 Spaces: 4 UTF-8 LF Java Properties
```